Carbon

Purpose

The carbon output family reports the state and fluxes of the SWAT+ Century-style carbon module: per-HRU and per-LSU gain and loss totals, soil C transformations, per-layer pool composition, plant carbon state, and process drivers. The family supersedes the old sprawl of 47 files in 26 base names. Every file is opt-in through `print.prt`: each carbon family has its own row with the standard four-slot `d / m / y / a` layout.

The outputs are produced only when `bsn_cc%cswat == 2` (the dynamic carbon model; the activation value moved from 1 to 2). See `codes.bsn` and the carbon model theory.

Source

- HRU writer: `src/hru_carbon_output.f90` (gain/loss and transformations) and `src/soil_nutcarb_write.f90` (per-layer pools, fluxes, drivers, dynamics, soil snapshot).
- LSU writer: `src/lsu_carbon_output.f90`.
- File opening and headers: `src/output_landscape_init.f90`.
- Header types and variable-name lists: `src/output_landscape_module.f90` (`output_carb_gl_header`, `output_hscf_header`, …); `src/carbon_module.f90` (`cpool_vars`, `n_p_pool_vars`, `cflux_vars`, `carb_drv_vars`, `carb_dyn_vars`, `soil_snap_vars`).
- Per-family `print.prt` parsing: `src/basin_print_codes_read.f90`, `hru_cb_*` and `lsu_cb_*` case branches.

File naming

Every file follows the canonical convention `<family>_{day,mon,yr,aa}.{txt,csv}`. The `_day` variant is daily, `_mon` is end-of-month, `_yr` is end-of-year, `_aa` is the average-annual record written once at end-of-simulation. The `hru_soil_snap` family uses `_tot` instead of `_aa` because an annual average of a snapshot pool is degenerate; the `_tot` file carries one begsim row and one endsim row, so `endsim_c - begsim_c` is the simulated SOC change per layer (see the carbon outputs reference).

CSV variants are emitted in parallel when `csvout = y` in `print.prt`. Both variants share the same column header and units row.

Per-family print.prt gating

Each carbon family has its own row in the per-object output block of `print.prt`. The `basin_print_codes_read.f90` `case (...)` table accepts these names; each row is one object label followed by four flags (`d m y a`):

`print.prt` row	Files gated	Output scope
`hru_cb_gl`	`hru_carb_gl_*.{txt,csv}`	HRU
`hru_cb_trf`	`hru_scf_*.{txt,csv}`	HRU
`hru_cb_lyr`	`hru_cbn_lyr_*.{txt,csv}`	HRU
`hru_cb_cpool`	`hru_cpool_stat_*.{txt,csv}`	HRU
`hru_cb_npool`	`hru_n_p_pool_stat_*.{txt,csv}`	HRU
`hru_cb_plt`	`hru_plc_stat_*.{txt,csv}`	HRU
`hru_cb_flux`	`hru_cflux_stat_*.{txt,csv}`	HRU
`hru_cb_drv`	`hru_carb_drv_*.{txt,csv}`	HRU
`hru_cb_dyn`	`hru_carb_dyn_*.{txt,csv}`	HRU
`hru_cb_snap`	`hru_soil_snap_*.{txt,csv}` (uses `_tot` instead of `_aa`)	HRU

print.prt row	Files gated	Output scope
lsu_cb_gl	lsu_carb_gl_*.{txt,csv}	LSU
lsu_cb_trf	lsu_scf_*.{txt,csv}	LSU
lsu_cb_plt	lsu_plc_stat_*.{txt,csv}	LSU

The output restructure split the previous coarse hru_cb flag into the ten per-family flags above. The old hru_cb and hru_cb_vars rows now drive a parallel set of legacy CSU output files (see below), reinstated so the CSU team's existing workflows keep working alongside the standardised per-family files.

The LSU files are emitted only when db_mx%lsu_out > 0 (the dataset contains routing units), regardless of the lsu_cb_* flag. Datasets without rout_unit.con populated do not create empty LSU carbon files.

Legacy CSU carbon outputs

A second, older set of carbon output files runs alongside the standardised per-family files above. These are not exposed in SWAT+ Editor, so regular users do not see them; the CSU team adds the driving rows to print.prt by hand. This legacy path is temporary and is marked for removal in revision 63.

Two print.prt rows drive it, both read by basin_print_codes_read.f90 and name-driven (a print.prt without them produces none of these files):

print.prt row	When	Files
hru_cb	always when present	hru_cbn_lyr, hru_seq_lyr, hru_n_p_pool_stat
hru_cb = l (layer detail)	flag letter l turns diagnostics on	adds hru_cpool_stat, hru_cflux_stat, hru_plc_stat, hru_begsim_soil_prop, hru_endsim_soil_prop, basin_carbon_all
hru_cb = y (profile)	flag letter y	light set only (diagnostics off)
hru_cb_vars	requires carbon on (cswat == 2)	hru_carbvars, hru_org_allo_vars, hru_org_ratio_vars, hru_org_trans_vars

Each file is written as .txt, plus .csv when csvout = y in print.prt.

The legacy and standardised files hold the same values in a different layout. For example, legacy hru_cbn_lyr (totals) plus hru_seq_lyr (sequestered) together equal the merged new hru_cbn_lyr_yr, which carries totals, sequestered, and depths in one wide table.

The internal cbn_diagnostics switch (which gates the heavy diagnostic files) is no longer read from an input file; it is set at startup from the hru_cb flag letter (l = on, y = off). Legacy output uses Fortran file units 8348-8387.

> **Note**
>
> The cbn_diagnostics flag in the old carb_coefs.cbn is retired. Output gating now lives entirely in print.prt.

HRU files

hru_carb_gl_* (family hru_cb_gl)

Merged HRU carbon gain and loss totals. 21 value columns covering soil C balance, residue C balance, and plant C balance. Replaces the former hru_soilcarb_*, hru_rescarb_*, and hru_plcarb_* triplet. Units are kg C/ha for every value column.

Columns (in declaration order in output_carb_gl_header): jday, mon, day, yr, unit, gis_id, name, then sed_c, surq_c, latq_c, perc_c, res_decay, man_app_c, man_graze_c, rsp_c, soil_emit_c, plant_surf_c, plant_root_c, rsd_surfdecay_c, rsd_rootdecay_c, harv_stov_c, rsd_emit_c, npp_c, harv_abgr_c, harv_root_c, drop_c, grazeat_c, plant_emit_c.

The output restructure broke out four sub-fields that the previous output silently summed under a single label: plant_surf_c, plant_root_c, harv_abgr_c, harv_root_c.

hru_scf_* (family hru_cb_trf)

HRU-level soil carbon transformations (the 13 inter-pool fluxes). One row per HRU per timestep.

Columns: jday, mon, day, yr, unit, gis_id, name, then meta_micr, str_micr, str_hs, co2_meta, co2_str, micr_hs, micr_hp, hs_micr, hs_hp, hp_micr, co2_micr, co2_hs, co2_hp.

hru_cbn_lyr_* (family hru_cb_lyr)

Per-layer total soil carbon, written wide (one row per HRU per timestep with per-layer columns). The number of layers in the output is cb_n_layers. When the optional carbon_layers.prt is absent this defaults to the largest soil layer count across all HRUs, so the deepest profile is written in full; when the file is present it fixes the count to the value given. Each row carries that HRU's own layer depths, and layers a soil does not have are filled with the sentinel -99.

Header is built at first data write and carries the structure layer depth (mm) | total_c_lyr1 | total_c_lyr2 | ... | total_c_lyrN. Units: Mg C/ha per layer column.

hru_cpool_stat_* (family hru_cb_cpool)

Per-layer C pools. Wide layout, with the columns in cpool_vars (src/carbon_module.f90:20) expanded to <var>_lyr1 ... <var>_lyrN:

residue_c, structural_c, metabolic_c, hs_c, hp_c, microbial_c,
lignin_c, nonlignin_c, root_mass, soil_water

10 variables x cb_n_layers columns. Units are kg C/ha for the C pools, kg/ha for root_mass, mm for soil_water.

hru_n_p_pool_stat_* (family hru_cb_npool)

Per-layer N and P content of the carbon pools. Variables in n_p_pool_vars:

tot_pool_n, residue_n, structural_n, metabolic_n, hs_n, hp_n,
microbial_n, lignin_n, nonlignin_n,
tot_pool_p, residue_p, structural_p, metabolic_p, hs_p, hp_p,
microbial_p, lignin_p, nonlignin_p

18 variables x cb_n_layers columns. Units kg N/ha or kg P/ha as the column name implies.

hru_plc_stat_* (family hru_cb_plt)

HRU-level plant carbon state. No per-layer expansion. Seven flat columns:

total_c, ab_gr_c, leaf_c, stem_c, seed_c, root_c, surf_rsd_c

Units kg C/ha.

hru_cflux_stat_* (family hru_cb_flux)

Per-layer C and N fluxes from the Century pool transformations. Variables in cflux_vars (37 entries):

```
cfmets1, cfstrs1, cfstrs2,                          (C flows from litter)
efmets1, efstrs1, efstrs2,                          (E = effective C)
immmets1, immstrs1, immstrs2,                       (immobilisation)
mnrmets1, mnrstrs1, mnrstrs2,                       (mineralisation)
co2fmet, co2fstr,                                   (CO2 from litter)
cfs1s2, cfs1s3, cfs2s1, cfs2s3, cfs3s1,             (inter-SOC pool C flows)
efs1s2, efs1s3, efs2s1, efs2s3, efs3s1,
imms1s2, imms1s3, imms2s1, imms2s3, imms3s1,
mnrs1s2, mnrs1s3, mnrs2s1, mnrs2s3, mnrs3s1,
co2fs1, co2fs2, co2fs3                              (CO2 from SOC pools)
```

37 variables x cb_n_layers columns. Pool indices: s1 = microbial, s2 = slow humus, s3 = passive humus, met = metabolic litter, str = structural litter.

hru_carb_drv_* (family hru_cb_drv)

Per-layer environmental drivers and intermediate factors for the Century transformations. Variables in carb_drv_vars (14 entries):

```
sut, tillagef, cons_bmix, tillagef_biomix, tillagef_tillmix,
till_eff, cdg, ox, cs, no3, nh4, co2_resp, soil_temp, emix
```

14 variables x cb_n_layers columns. sut, cdg, ox, cs are dimensionless factors; no3, nh4 are kg/ha; co2_resp is kg C/ha; soil_temp is deg C; tillagef*, till_eff, cons_bmix, emix are dimensionless.

hru_carb_dyn_* (family hru_cb_dyn)

Per-layer carbon pool dynamics: allocation fractions and the C-to-N ratios of each pool transformation. Variables in carb_dyn_vars (21 entries):

```
asp, abp, abco2, a1co2, asco2, apco2,              (CO2 / passive allocations)
ncbm, nchp, nchs,                                  (N:C of microbial, passive, slow)
bmctp, bmntp, hsctp, hsntp, hpctp, hpntp,          (pool C and N totals)
lmctp, lmntp, lsctp, lslctp, lslnctp, lsntp        (litter pool totals)
```

21 variables x cb_n_layers columns. Allocation fractions are dimensionless; pool totals are kg C/ha or kg N/ha.

hru_soil_snap_* (family hru_cb_snap)

Per-layer soil snapshot. Uses _tot in place of _aa because an annual average of a snapshot is degenerate.

Variables in soil_snap_vars (13 entries):

```
bd, awc, soil_k, tot_c, clay, silt, sand, rock,
alb, usle_k, ec, caco3, ph
```

13 variables x cb_n_layers columns.

The _day, _mon, _yr files emit end-of-period rows on the usual cadence. The _tot file is special: it emits one row at simulation begin (begsim) and one at simulation end (endsim). Both rows carry pool values in the same unit (kg C/ha for tot_c), so endsim_tot_c_lyrK - begsim_tot_c_lyrK is the simulated SOC change for layer K. The previous design wrote a percentage on the begsim row and a mass on the endsim row, which made the delta meaningless.

LSU files

The LSU files are area-weighted HRU aggregates produced by lsu_carbon_output.f90. Per-layer families are **not** aggregated to LSU because different HRUs within an LSU can have different soil profiles.

lsu_carb_gl_* (family lsu_cb_gl)

LSU-level carbon gain and loss totals, area-weighted from the HRU hru_carb_gl_* values. Reuses the output_carb_gl_header columns (same 21 value columns as the HRU file).

lsu_scf_* (family lsu_cb_trf)

LSU-level soil C transformations, area-weighted from the HRU values. Reuses the output_hscf_header columns.

lsu_plc_stat_* (family lsu_cb_plt)

LSU-level plant carbon, summed across HRUs with the ru_frac weight. Single value column plant_c (kg C/ha), printed once per LSU per timestep.

Optional supporting file

`carbon_layers.prt`

Optional file that fixes `cb_n_layers`, the number of soil layers expanded in every wide per-layer carbon file. It is only needed when you want a specific column count; if it is absent the model uses the largest soil layer count across all HRUs (see below). Format:

- Line 1: title (free text, skipped).
- Line 2: header (free text, skipped).
- Line 3: single integer giving the layer count.

Only line 3 carries data. The reader skips lines 1 and 2 entirely, so their text is for the reader's own benefit and may say anything. An integer below 1 is rejected and the automatic default is used instead.

Example

```
carbon_layers.prt: layers to expand in per-layer carbon output
n_layers
10
```

Here line 1 is a free-text title, line 2 is a free-text header, and line 3 is the only value the reader uses. With `10`, every per-layer carbon file carries ten layer columns regardless of the soils present, and any unused slots are filled by `-99`. Use this to force a fixed column count, for example to truncate output to the top layers or to keep a constant width across runs.

If the file is absent, `cb_n_layers` defaults to the largest soil layer count across all HRUs. This means the deepest profile is written in full and no real layer is truncated; HRUs with fewer layers than that maximum get the sentinel `-99` in their extra slots. (Earlier versions defaulted to a fixed 7, which silently truncated deeper soils.) Reader: `src/carbon_layers_read.f90`. Not listed in `file.cio`.

Related pages

- Carbon (model theory). Pool definitions and Century-style transformation equations.
- `carbon.bsn`. Basin-wide scalar inputs.
- `carbon_lyr.bsn`. Per-layer inputs.
- `print.prt`. Output gating.
- Naming convention. Common naming rules across output files.